INCH-POUND

MS16624E
21 July 2022
SUPERSEDING
MS16624D
w/AMENDMENT 2
26 August 2013

DETAIL SPECIFICATION SHEET

RING, RETAINING, EXTERNAL, BASIC (TAPERED SECTION TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-DTL-21248.

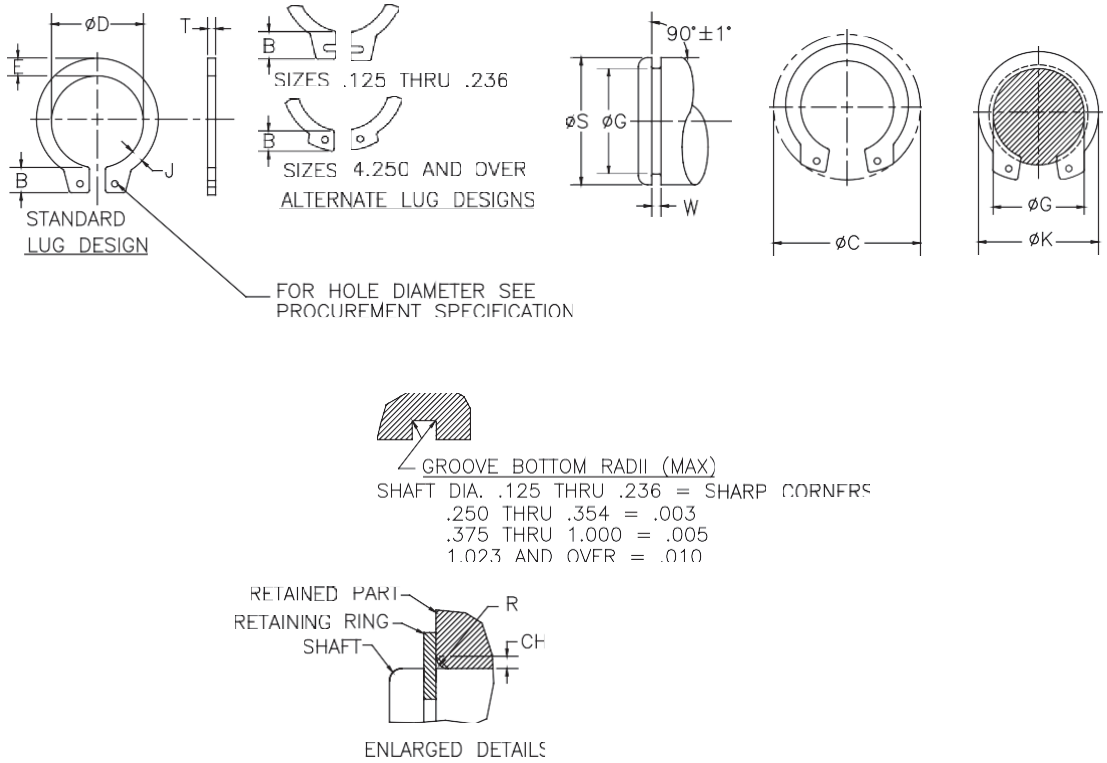


FIGURE 1. Ring, Retaining.

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency - Aviation, ATTN: DLA Aviation-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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TABLE I. Dimensions.

ØS SHAFT (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION* HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH RECOMMENDED GROOVE (REF)		ØK 3/	ØC 4/ CLEAR	R 5/ OF RETAINED PART (REF)	CH 5/ OF RETAINED PART (REF)
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX
.125	3.2	.112		.046		.018	±.0015	.011	±.0015	.010	±.001	.117		.012		.148	.222	.010	.006
.156	4.0	.142		.054		.026		.016		.010		.146		.012		.189	.270	.015	.009
.188	4.8	.168		.050		.025		.016		.015		.175	±.0015	.018	+.002 -.000	.218	.298	.014	.008
			+.002		±.002														
.197	5.0	.179	-.004	.056		.026	±.002	.016	±.002	.015		.185	.0015	.018		.229	.319	.014	.009
.219	5.6	.196		.056		.028		.017		.015		.205	FIM 2/	.018		.252	.338	.015	.009
.236	6.0	.215		.056		.030		.019		.015		.222		.018		.272	.355	.016	.010
.250	6.4	.225		.080		.035		.025		.025		.230		.030		.290	.450	.018	.011
.276	7.0	.250		.081		.035		.024		.025		.255		.030		.315	.480	.017	.010
.281	7.1	.256		.080		.038		.025		.025		.261		.030		.326	.490	.020	.012
.312	7.9	.281		.087		.040		.026		.025		.290		.030		.357	.540	.020	.012
.344	8.7	.309		.087		.042	±.003	.026	±.003	.025		.321	±.002	.030		.390	.570	.021	.012
.354	9.0	.320	+.002 -.005	.087		.046		.029		.025	±.002	.330	.002	.030		.405	.590	.023	.014
.375	9.5	.338		.088		.050		.030		.025		.352	FIM 2/	.030		.433	.610	.026	.015
.394	10.0	.354		.087	±.003	.052		.031		.025		.369		.030	+.003	.452	.620	.027	.016
.406	10.3	.366		.087		.054		.033		.025		.382		.030	-.000	.468	.630	.028	.017
.438	11.1	.395		.088		.055		.033		.025		.412		.030		.501	.660	.029	.017
.469	11.9	.428		.088		.060		.035		.025		.443		.030		.540	.680	.031	.018
.500	12.7	.461		.108		.065		.040		.035		.468	±.002	.040		.574	.770	.034	.020
.551	14.0	.509		.108		.053		.036		.035		.519	.004	.040		.611	.810	.027	.016
.562	14.3	.521		.108		.072	±.004	.041	±.004	.035		.530	FIM 2/	.040		.644	.820	.038	.023
.594	15.1	.550	+.005 -.010	.109		.076		.043		.035		.559		.040		.680	.860	.039	.023
.625	15.9	.579		.110		.080		.045		.035		.588	±.003	.040		.715	.900	.041	.025
.669	17.0	.621		.110		.082		.043		.035		.629	.004	.040		.756	.930	.040	.024
.672	17.1	.621		.110		.082		.043		.035		.631	FIM 2/	.040		.758	.930	.040	.024

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TABLE I. Dimensions – Continued.

ØS SHAFT (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION* HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		ØK 3/	ØC 4/ CLEAR	R 5/ OF RETAINED PART (REF)	CH 5/
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX
.688	17.5	.635		.136		.084		.048		.042		.646		.047		.779	1.010	.042	.025
.750	19.0	.693		.136		.092		.051		.042		.704		.047		.850	1.090	.046	.027
.781	19.8	.722		.136		.094		.052		.042		.733	±.003	.047		.883	1.120	.047	.028
															+.003				
.812	20.6	.751	+.005	.136		.096	±.005	.054	±.005	.042		.762		.047	-.000	.914	1.150	.047	.028
.875	22.2	.810	-.010	.137		.104		.057		.042		.821	.004	.047		.987	1.210	.051	.030
.938	23.8	.867		.166		.110		.063		.042		.882	FIM 2/	.047		1.054	1.340	.055	.033
.984	25.0	.910		.167		.114		.064		.042		.926		.047		1.106	1.390	.056	.033
1.000	25.4	.925		.167		.116		.065		.042	±.002	.940		.047		1.122	1.410	.057	.034
1.023	26.0	.946		.168		.118		.066		.042		.961		.047		1.147	1.430	.058	.035
1.062	27.0	.982		.181		.122		.069		.050		.998		.056		1.192	1.500	.060	.036
1.125	28.6	1.041		.182		.128		.071		.050		1.059		.056		1.261	1.550	.063	.038
1.188	30.2	1.098		.182		.132		.072		.050		1.118	±.004	.056		1.325	1.610	.064	.038
			+.010																
1.250	31.7	1.156	-.015	.183	±.004	.140		.076		.050		1.176	.005	.056		1.396	1.690	.068	.041
1.312	33.3	1.214		.183		.146		.076		.050		1.232	FIM 2/	.056		1.458	1.750	.068	.041
1.375	34.9	1.272		.184		.152		.082		.050		1.291		.056		1.529	1.800	.072	.043
1.438	36.5	1.333		.184		.160		.086		.050		1.350		.056	+.004	1.600	1.870	.076	.045
1.500	38.1	1.387		.214		.168	±.006	.091	±.006	.050		1.406		.056	-.000	1.668	1.990	.079	.047
1.562	39.7	1.446		.235		.172		.093		.062		1.468		.068		1.740	2.100	.082	.049
1.625	41.3	1.503		.235		.180		.097		.062		1.529		.068		1.812	2.170	.087	.052
1.688	42.9	1.560		.235		.184		.099		.062		1.589	±.005	.068		1.877	2.240	.090	.054
1.750	44.4	1.618	+.013	.237		.188		.101		.062		1.650		.068		1.945	2.310	.091	.054
			-.020										.005						
1.772	45.0	1.637		.237		.190		.102		.062		1.669	FIM 2/	.068		1.967	2.330	.092	.055
1.812	46.0	1.675		.238		.192		.102		.062		1.708		.068		2.010	2.380	.092	.055
1.875	47.6	1.735		.239		.196		.104		.062		1.769		.068		2.076	2.440	.094	.056
1.969	50.0	1.819		.245		.200		.106		.062		1.857		.068		2.170	2.540	.094	.056
2.000	50.8	1.850		.239		.204		.108		.062		1.886		.068		2.205	2.550	.096	.057
2.062	52.4	1.906		.266		.208		.111		.078		1.946		.086		2.275	2.680	.098	.059
2.125	54.0	1.964		.266		.212		.113		.078	±.003	2.003		.086		2.337	2.750	.098	.059
2.156	54.8	1.993		.266		.212		.113		.078		2.032		.086		2.366	2.780	.097	.058
2.250	57.1	2.081	+.015	.267		.220		.116		.078		2.120		.086		2.466	2.870	.100	.060
			-.025																
2.312	58.7	2.139		.267		.222		.118		.078		2.178	±.006	.086		2.528	2.940	.100	.060
2.375	60.3	2.197		.267		.224		.119		.078		2.239		.086	+.005	2.591	3.010	.100	.060
2.438	61.9	2.255		.268	±.005	.228	±.007	.120	±.007	.078		2.299	.006 FIM 2/	.086	-.000	2.657	3.070	.102	.061
2.500	63.5	2.313		.268		.232		.122		.078		2.360		.086		2.724	3.120	.104	.062
2.559	65.0	2.377		.268		.238		.125		.078		2.419		.086		2.792	3.180	.108	.065
2.625	66.7	2.428		.268		.242		.127		.078		2.481		.086		2.860	3.250	.109	.066
2.688	68.3	2.485	+.020	.268		.246		.129		.078		2.541		.086		2.926	3.320	.111	.067
2.750	69.8	2.543	-.030	.310		.248		.131		.093		2.602		.103		2.992	3.450	.112	.067*
2.875	73.0	2.659		.308		.256		.133		.093		2.721		.103		3.122	3.570	.115	.069

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TABLE I. Dimensions – Continued.

ØS SHAFT (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION* HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		ØK 3/	ØC 4/ CLEAR	R 5/ OF RETAINED PART (REF)	CH 5/
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX
2.938	74.6	2.717		.308		.260	±.007	.136	±.007	.093		2.779		.103		3.187	3.640	.116	.070
3.000	76.2	2.775		.308		.264		.138		.093		2.838		.103		3.252	3.690	.117	.070
3.062	77.8	2.832		.298		.252		.131		.093		2.898		.103		3.294	3.740	.107	.064
3.125	79.4	2.892		.308		.272		.141		.093		2.957		.103		3.383	3.820	.120	.072
3.156	80.2	2.920		.308		.274		.143		.093		2.986		.103		3.415	3.850	.120	.072
3.250	82.0	3.006		.308	±.005	.280	±.008	.145		.093		3.076		.103		3.515	3.950	.123	.074
3.346	85.0	3.092		.308		.286		.147		.093		3.166		.103		3.613	4.040	.126	.076
3.438	87.3	3.179	+.020	.308		.292		.148	±.008	.093		3.257		.103		3.712	4.140	.129	.077
3.500	88.9	3.237	-.030	.328		.285		.148		.109		3.316		.120		3.764	4.250	.122	.073
3.543	90.0	3.277*		.328		.288		.149		.109	±.003	3.357	±.006	.120	+.005	3.809	4.290	.123	.074
3.625	92.0	3.352		.328		.296		.153		.109		3.435	.006	.120	-.000	3.898	4.370	.127	.076
3.688	93.7	3.410		.330		.302		.156		.109		3.493	FIM 2/	.120		3.966	4.430	.129	.078
3.750	95.2	3.468		.332		.310		.160		.109		3.552		.120		4.037	4.500	.133	.080
3.875	98.4	3.584		.330		.318		.163		.109		3.673		.120		4.169	4.600	.137	.082
3.938	100.0	3.642		.342		.318		.163		.109		3.734		.120		4.230	4.700	.137	.082
4.000	101.6	3.700		.352		.318		.163		.109		3.792		.120		4.288	4.780	.135	.081
4.250	108.0	3.989		.395		.318		.176		.109		4.065		.120		4.558	5.090	.146	.088
4.375	111.1	4.106		.395		.318		.181		.109		4.190		.120		4.683	5.220	.146	.088
4.500	114.3	4.223		.404		.285		.128		.109		4.310		.120		4.730	5.370	.102	.061
4.750	120.6	4.458		.429		.303		.136		.109		4.550		.120		4.996	5.670	.115	.069
5.000	127.0	4.692		.450	±.008	.360	±.010	.194	±.010	.109		4.790		.120		5.346	5.960	.165	.099
5.250	133.3	4.927		.472		.372		.211		.125		5.030	±.007	.139	+.006	5.605	6.270	.169	.101
5.500	139.7	5.162	+.020	.497		.390		.209		.125	±.004	5.265		.139	-.000	5.867	6.570	.175	.105
5.750	146.0	5.396	-.040	.518		.408		.220		.125		5.505	.006 FIM 2/	.139		6.134	6.860	.184	.110
6.000	152.4	5.631		.540		.381		.171		.125		5.745		.139		6.302	7.160	.143	.086
6.250	158.7	5.866		.561		.396		.176		.156		5.985		.174		6.568	7.460	.148	.089
6.500	165.1	6.100	+.020 -.050	.586		.438		.236		.156		6.225		.174		6.905	7.870	.191	.114
6.750	171.4	6.335		.608		.456		.246		.156		6.465	±.008	.174	+.008	7.172	8.060	.200	.120
7.000	177.8	6.570		.530		.474		.256		.156		6.705		.174	-.000	7.439	8.360	.208	.125
7.500	190.5	7.009		.676	±.012	.507	±.015	.277	±.015*	.187	±.005	7.180	.006 FIM 2/	.209		7.963	8.960	.220	.132
8.000	203.2	7.478		.735		.540		.294		.187		7.660		.209		8.493	9.600	.235	.141
8.500	215.9	7.947	+.050	.735		.573		.314		.187		8.140		.209		9.023	10.100	.250	.150
9.000	228.6	8.415	-.130	.735		.609		.333		.187		8.620		.209		9.557	10.600	.267	.160
9.500	241.3	8.885		.735		.642		.350		.187		9.100		.209		10.086	11.100	.281	.168
10.00	254.0	9.355		.735		.675		.367		.187		9.575		.209		10.610	11.600	.294	.176

1/ T - THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.

2/ FIM - (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.

3/ K - MAXIMUM DIAMETER WHEN THE RING IS PROPERLY SEATED IN THE GROOVE. (DESIGN REFERENCE DIMENSION).

4/ C - ACTUAL CLEARANCE DIAMETER WHEN THE RING IS SPRUNG OVER THE SHAFT PRIOR TO INSTALLATION INTO THE GROOVE. (DESIGN REFERENCE DIMENSION).

5/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. ALLOWABLE THRUST LOADS FOR RINGS RETAINING PARTS WITH MAXIMUM CORNER RADII OR CHAMFERS ARE LISTED IN TABLE VII ON PAGE 11.

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I, CLASS 2 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL: CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568/A568M OR ASTM A684/A684M.
CORRISION RESISTANT STEEL, IN ACCORDANCE WITH SAE AMS5520 (UNS S15700).
BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194, HEAT TREAT TO TEMPER DESIGNATION CODE TO MEET HARDNESS REQUIREMENTS IN TABLE II. BERYLLIUM COPPER ALLOYS SHALL NOT BE SPECIFIED FOR GROUND VEHICLE SYSTEMS.

3. HARDNESS

TABLE II. Hardness.

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.125 TO .236			77 TO 82 HRI5N
.250 TO .669	68.5-73.0 HR30N		
.668 TO 1.500	52.5-59.0 HR45N		
1.562 TO 3.437	46.0-51.0 HRC		
3.500 & ABOVE	44.0-49.0 HRC		
.250 TO .812		63-69.5 HR30N	
.875 & ABOVE		44-51 HRC	
.250 TO 1.023			56.5 TO 62 HR30N
1 .062 & ABOVE			37 TO 43 HRC

4. PROTECTIVE FINISH OR TREATMENT:

a. CARBON STEEL - SHALL BE (SEE TABLE III, IV, OR V)

1. CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5. CADMIUM IS AN R1, R2, AND D1 HAZARDOUS SUBSTANCE IN THE AEROSPACE AND DEFENSE DECLARABLE SUBSTANCE LIST. DO NOT USE THIS DESIGNATOR FOR NEW DESIGNS. FOR MOST APPLICATIONS, ZINC WITH HEXAVALENT CHROMIUM-FREE PASSIVATION AS SPECIFIED IN 4.a.4 IS A SUITABLE ALTERNATIVE. HOWEVER, THE USER MAY CONSIDER A CORROSION RESISTANT STEEL IF GREATER CORROSION RESISTANCE IS DESIRED.

2. ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn 5 or ASTM B695, TYPE II, CLASS 5. TYPE II CONTAINS HEXAVALENT CHROMIUM, WHICH IS AN R1, R2, AND D1 HAZARDOUS SUBSTANCE IN THE AEROSPACE AND DEFENSE DECLARABLE SUBSTANCE LIST. DO NOT USE THIS DESIGNATOR FOR NEW DESIGNS. FOR MOST APPLICATIONS, ZINC WITH HEXAVALENT CHROMIUM-FREE PASSIVATION AS SPECIFIED IN 4.a.4 IS A SUITABLE ALTERNATIVE.

3. PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS 2.

4. ZINC COATING OPTIONS WITH HEXAVALENT CHROMIUM-FREE PASSIVATION. MANUFACTURER MAY CHOOSE FROM THE FOLLOWING (SEE NOTES SECTION, ITEM 4):

- ZINC WITH HEXAVALENT CHROMIUM-FREE PASSIVATION MECHANICALLY APPLIED IN ACCORDANCE WITH ASTM B695, TYPE II, EXCEPT THE SUPPLEMENTARY TREATMENT SHALL BE HEXAVALENT CHROMIUM-FREE.
- ZINC COAT IN ACCORDANCE WITH ASTM B633, CLASS FE/ZN5, TYPE VI, COLOR OPTIONAL, WITH HEXAVALENT CHROMIUM-FREE PASSIVATE. BAKING FOR HYDROGEN EMBRITTLEMENT RELIEF IS MANDATORY.

b. CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH METHOD 1, TYPE 8, OR METHOD 2, SAE AMS2700, OR TREATMENT NITRIC 4, OR CITRIC 1 THROUGH 5, ASTM A967/A967M, EXCEPT HEXAVALENT CHROMIUM CONTAINING PROCESSES ARE NOT PERMITTED.

c. BERYLLIUM COPPER - CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3, OR ASTM B696, TYPE II, CLASS 5.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III, IV, OR V.

EXAMPLE: MS16624-6100 IS THE PART NUMBER FOR A CARBON STEEL, ZINC/HEX-FREE PASSIVATE, EXTERNAL BASIC RETAINING RING FOR USE ON A 1.000 SHAFT DIAMETER.

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TABLE III. Dash numbers for MS16624.

Ø S SHAFT (REF)	1/ 2/ CARBON STEEL CAD PLATE	1/ 2/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSP HATE COAT	STEEL, CORRO-SION RESIS-TANT (CRES)	1/ Be COPPER	1/ 2/ Be COPPER CAD PLATE	1/ CARBON STEEL ZINC/HEX FREE PASSIVATE
	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
.125					5012	5012-1	6012
.156					5015	5015-1	6015
.188					5018	5018-1	6018
.197					5019	5019-1	6019
.219					5021	5021-1	6021
.236					5023	5023-1	6023
.250	1025	2025	3025	4025	5025	5025-1	6025
.276	1027	2027	3027	4027	5027	5027-1	6027
.281	1028	2028	3028	4028	5028	5028-1	6028
.312	1031	2031	3031	4031	5031	5031-1	6031
.344	1034	2034	3034	4034	5034	5034-1	6034
.354	1035	2035	3035	4035	5035	5035-1	6035
.375	1037	2037	3037	4037	5037	5037-1	6037
.394	1039	2039	3039	4039	5039	5039-1	6039
.406	1040	2040	3040	4040	5040	5040-1	6040
.438	1043	2043	3043	4043	5043	5043-1	6043
.469	1046	2046	3046	4046	5046	5046-1	6046
.500	1050	2050	3050	4050	5050	5050-1	6050
.551	1055	2055	3055	4055	5055	5055-1	6055
.562	1056	2056	3056	4056	5056	5056-1	6056
.594	1059	2059	3059	4059	5059	5059-1	6059
.625	1062	2062	3062	4062	5062	5062-1	6062
.669	1066	2066	3066	4066	5066	5066-1	6066
.672	1067	2067	3067	4067	5067	5067-1	6067
.688	1068	2068	3068	4068	5068	5068-1	6068
.750	1075	2075	3075	4075	5075	5075-1	6075
.781	1078	2078	3078	4078	5078	5078-1	6078
.812	1081	2081	3081	4081	5081	5081-1	6081
.875	1087	2087	3087	4087	5087	5087-1	6087
.938	1093	2093	3093	4093	5093	5093-1	6093
.984	1098	2098	3098	4098	5098	5098-1	6098
1.000	1100	2100	3100	4100	5100	5100-1	6100
1.023	1102	2102	3102	4102	5102	5102-1	6102

Ø S SHAFT (REF)	1/ 2/ CARBON STEEL CAD PLATE	1/ 2/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL, CORRO-SION RESIS-TANT (CRES)	1/ Be COPPER	1/ 2/ Be COPPER CAD PLATE	1/ CARBON STEEL ZINC/HEX FREE PASSIVATE
	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
1.062	1106	2106	3106	4106	5106	5106-1	6106
1.125	1112	2112	3112	4112	5112	5112-1	6112
1.188	1118	2118	3118	4118	5118	5118-1	6118
1.250	1125	2125	3125	4125	5125	5125-1	6125
1.312	1131	2131	3131	4131	5131	5131-1	6131
1.375	1137	2137	3137	4137	5137	5137-1	6137
1.438	1143	2143	3143	4143	5143	5143-1	6143
1.500	1150	2150	3150	4150	5150	5150-1	6150
1.562	1156	2156	3156	4156	5156	5156-1	6156
1.625	1162	2162	3162	4162	5162	5162-1	6162
1.688	1168	2168	3168	4168	5168	5168-1	6168
1.750	1175	2175	3175	4175	5175	5175-1	6175
1.772	1177	2177	3177	4177	5177	5177-1	6177
1.812	1181	2181	3181	4181	5181	5181-1	6181
1.875	1187	2187	3187	4187	5187	5187-1	6187
1.969	1196	2196	3196	4196	5196	5196-1	6196
2.000	1200	2200	3200	4200	5200	5200-1	6200
2.062	1206	2206	3206	4206			6206
2.125	1212	2212	3212	4212			6212
2.156	1215	2215	3215	4215			6215
2.250	1225	2225	3225	4225			6225
2.312	1231	2231	3231	4231			6231
2.375	1237	2237	3237	4237			6237
2.438	1243	2243	3243	4243			6243
2.500	1250	2250	3250	4250			6250
2.559	1255	2255	3255	4255			6255
2.625	1262	2262	3262	4262			6262
2.688	1268	2268	3268	4268			6268
2.750	1275	2275	3275	4275			6275
2.875	1287	2287	3287	4287			6287
2.938	1293	2293	3293	4293			6293
3.000	1300	2300	3300	4300			6300
3.062	1306	2306	3306	4306			6306

1/ Substitute corrosion resistant steel when used in food processing machinery, or in fuel or lubrication systems, or when used at temperatures over 450°F (232°C).

2/ See restricted substance note 5.

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TABLE III. Dash numbers for MS16624 – Continued.

Ø S SHAFT (REF)	1/ 2/ CARBON STEEL CAD PLATE	1/ 2/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL CORROSION RESISTANT (CRES)	1/ Be COPPER	1/ 2/ Be COPPER CAD PLATE	CARBON STEEL ZINC/HEX FREE PASSIVATE
	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
3.125	1312	2312	3312	4312			6312
3.156	1315	2315	3315	4315			6315
3.250	1325	2325	3325	4325			6325
3.346	1334	2334	3334	4334			6334
3.438	1343	2343	3343	4343			6343
3.500	1350	2350	3350	4350			6350
3.543	1354	2354	3354	4354			6354
3.625	1362	2362	3362	4362			6362
3.688	1368	2368	3368	4368			6368
3.750	1375	2375	3375	4375			6375
3.875	1387	2387	3387	4387			6387
3.938	1393	2393	3393	4393			6393
4.000	1400	2400	3400	4400			6400
4.250	1425	2425	3425	4425			6425
4.375	1437	2437	3437	4437			6437
4.500	1450	2450	3450	4450			6450
4.750	1475	2475	3475	4475			6475
5.000	1500	2500	3500	4500			6500
5.250	1525	2525	3525	4525			6525
5.500	1550	2550	3550	4550			6550
5.750	1575	2575	3575	4575			6575
6.000	1600	2600	3600	4600			6600
6.250	1625	2625	3625	4625			6625
6.500	1650	2650	3650	4650			6650
6.750	1675	2675	3675	4675			6675
7.000	1700	2700	3700	4700			6700
7.500	1750	2750	3750	4750			6750
8.000	1800	2800	3800	4800			6800
8.500	1850	2850	3850	4850			6850
9.000	1900	2900	3900	4900			6900
9.500	1950	2950	3950	4950			6950
10.000	2000	3000	4000	5000			7000

1/ Substitute corrosion resistant steel when used in food processing machinery, or in fuel or lubrication systems, or when used at temperatures over 450°F (232°C).

2/ See restricted substance note 5.

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TABLE IV. Substitution table, cross reference part numbers.

Ø S SHAFT (REF)	INACTIVE CARBON STEEL MS9013	INACTIVE CARBON STEEL MS16624	SUBST 1/ CARBON STEEL ZINC/HEX FREE PASSIVATE MS16624	SUBST 1/ CARBON STEEL PHOSPHATE COAT MS16624	Ø S SHAFT (REF)	INACTIVE CARBON STEEL MS9013	INACTIVE CARBON STEEL MS16624	SUBST 1/ CARBON STEEL ZINC/HEX FREE PASSIVATE MS16624	SUBST 1/ CARBON STEEL PHOSPHATE COAT MS16624	Ø S SHAFT (REF)	INACTIVE CARBON STEEL MS16624	SUBST 1/ CARBON STEEL ZINC/HEX FREE PASSIVATE MS16624	SUBST 1/ CARBON STEEL PHOSPHATE COAT MS16624
.250	01	25	6025	3025	1.250	30	125	6125	3125	3.125	312	6312	3312
.276	02	27	6027	3027	1.312	31	131	6131	3131	3.156	315	6315	3315
.281	03	28	6028	3028	1.375	32	137	6137	3137	3.250	325	6325	3325
.312	04	31	6031	3031	1.438	33	143	6143	3143	3.346	334	6334	3334
.344	05	34	6034	3034	1.500	34	150	6150	3150	3.438	343	6343	3343
.354	06	35	6035	3035	1.562	35	156	6156	3156	3.500	350	6350	3350
.375	07	37	6037	3037	1.625	36	162	6162	3162	3.543*	354	6354	3354
.394	08	39	6039	3039	1.688	37	168	6168	3168	3.625	362	6362	3362
.406	09	40	6040	3040	1.750	38	175	6175	3175	3.688	368	6368	3368
.438	10	43	6043	3043	1.772	39	177	6177	3177	3.750	375	6375	3375
.469	11	46	6046	3046	1.812	40	181	6181	3181	3.875	387	6387	3387
.500	12	50	6050	3050	1.875	41	187	6187	3187	3.938	393	6393	3393
.551	13	55	6055	3055	1.969	42	196	6196	3196	4.000	400	6400	3400
.562	14	56	6056	3056	2.000	43	200	6200	3200	4.250	425	6425	3425
.594	15	59	6059	3059	2.062	44	206	6206	3206	4.375	437	6437	3437
.625	16	62	6062	3062	2.125	45	212	6212	3212	4.500	450	6450	3450
.669	17	66	6066	3066	2.156	46	215	6215	3215	4.750	475	6475	3475
.672		67	6067	3067	2.250	47	225	6225	3225	5.000	500	6500	3500
.688	18	68	6068	3068	2.312	48	231	6231	3231	5.250	525	6525	3525
.750	19	75	6075	3075	2.375	49	237	6237	3237	5.500	550	6550	3550
.781	20	78	6078	3078	2.438	50	243	6243	3243	5.750	575	6575	3575
.812	21	81	6081	3081	2.500	51	250	6250	3250	6.000	600	6600	3600
.875	22	87	6087	3087	2.559	52	255	6255	3255	6.250	625	6625	3625
.938	23	93	6093	3093	2.625	53	262	6262	3262	6.500	650	6650	3650
.984	24	98	6098	3098	2.688	54	268	6268	3268	6.750	675	6675	3675
1.000	25	100	6100	3100	2.750	55	275	6275	3275	7.000	700	6700	3700
1.023	26	102	6102	3102	2.875	56	287	6287	3287	7.500	750	6750	3750
1.062	27	106	6106	3106	2.938	57	293	6293	3293	8.000	800	6800	3800
1.125	28	112	6112	3112	3.000	58	300	6300	3300	8.500	850	6850	3850
1.188	29	118	6118	3118	3.062		306	6306	3306	9.000	900	6900	3900
										9.500	950	6950	3950
										10.000	1000	7000	4000

1/ Substitute corrosion resistant steel when used in food processing machinery, or in fuel or lubrication systems, or when used at temperatures over 450°F (232°C).

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TABLE V. Substitution table (cross reference of part numbers).

ØS SHAFT (REF)	INACTIVE	INACTIVE	SUBST
	BERYLLIUM 1/ COPPER	BERYLLIUM 1/ COPPER	BERYLLIUM 1/ COPPER
	MS9012	MS16624	MS16624
.125	01	12	5012
.156	02	15	5015
.188	03	18	5018
.197	04	19	5019
.219	05	21	5021
.236	06	23	5023

ØS SHAFT (REF)	INACTIVE	SUBST
	BERYLLIUM 1/ COPPER CAD* PLATE	BERYLLIUM 1/ COPPER CAD* PLATE
	MS16624	MS16624
.125	1012	5012-1
.156	1015	5015-1
.188	1018	5018-1
.197	1019	5019-1
.219	1021	5021-1
.236	1023	5023-1

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (232°C).

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. RECOMMENDED DESIGN LIMITATIONS AND USAGE.

(a) INTENDED USE- TO PROVIDE SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON SHAFTS. TAPERED DESIGN PRINCIPLE PERMITS RINGS TO MAINTAIN PRACTICALLY CONSTANT CIRCULARITY AND PRESSURE AGAINST BOTTOM OF GROOVE, COUNTERACTING CONSIDERABLE CENTRIFUGAL FORCE (SEE TABLE IV). RINGS FOR SHAFT DIAMETERS OVER 4 INCHES ARE SPECIALLY DIMENSIONED TO MAINTAIN BALANCE IN ROTATION. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE- THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED (PROVIDING GROOVE HAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

FOR APPROXIMATE SAFETY RPM LIMITS, SEE TABLE VI.

TABLE VI. Approximate safety RPM limits.

Ø SHAFT(INCHES)		.125	.500	1	2	3	4	6	8	10
CARBON AND CORROSION RESISTANT STEEL	RPM LIMIT	80,000	40,000	20,000	10,000	6,700	5,000	3,400	2,500	2,000
BERYLLIUM COPPER	RPM LIMIT	50,000	25,000	13,000	6,400					

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(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS. ABUTTING COMPONENTS TO HAVE SHARP CORNERS=

$$P = \frac{\pi STX}{F}$$

WHERE:
P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A SAFETY FACTOR, F= 4, IS RECOMMENDED SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO THE PURE SHEAR STRESSES, BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL

$$P = \frac{\pi SdY}{F}$$

WHERE:
P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO ENSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F= 2, IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT =

$$Z = 3d$$

WHERE:
Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE AND END OF SHAFT (INCHES)
d = GROOVE DEPTH {INCHES)

(e) DIFFERENTIAL ROTATION =

THE CONDITIONS UNDER WHICH A RETAINING RING MAY BE USED WHEN ADJACENT PARTS ROTATE RELATIVE TO IT, FALL INTO TWO CATEGORIES:

1. WHERE NO THRUST IS EXERTED BY ADJACENT PART; IN THIS CASE, DIFFERENTIAL ROTATION OF RING AND ADJACENT PART CREATES NO ELEMENT OF RISK IN THE APPLICATION OF THE RINGS BECAUSE NO FRICTIONAL TORQUE IS EXERTED BY THE MACHINE PART ON THE RING.

2. CONSIDERATION MUST BE GIVEN TO THE MAGNITUDE OF THE THRUST INVOLVED. THE FRICTION MOMENT MAY NOT EXCEED THE BENDING MOMENT, WHICH THE RING CAN TOLERATE WITHOUT RELEASING ITS PRESSURE AGAINST THE BOTTOM OF THE GROOVE, FORMULATED AS FOLLOWS:

$$fPN \leq \frac{sTE^2}{18} \quad \text{OR} \quad P \leq \frac{sTE^2}{f18N}$$

WHERE:
P = ALLOWABLE THRUST LOAD EXERTED BY ADJACENT PART (POUNDS)
f = COEFFICIENT OF FRICTION
s = WORKING STRESS OF RING UNDER MAXIMUM EXPANSION (PSI) 2/
T = RING THICKNESS (INCHES)
E = GREATEST WIDTH SECTION OF RING (INCHES)
N = NEUTRAL RING DIAMETER (INCHES), FREE DIAMETER PLUS 3/4 E DIMENSION

IN SUCH CASES WHERE DIFFERENTIAL ROTATION OCCURS, THE CALCULATION SHOULD BE BASED ON THE MAXIMUM POSSIBLE VALUE OF THE COEFFICIENT OF FRICTION.

1/ X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .672 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .688 INCH AND OVER SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL SHAFT DIAMETERS AND OF BERYLLIUM COPPER.
2/ S = 250,000 PSI WORKING STRESS FOR RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
S = 200,000 PSI WORKING STRESS FOR RINGS OF BERYLLIUM COPPER.

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(f) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ - FOR THE RING (INCH POUNDS), ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$I_G = \frac{Pd}{2}$ - FOR THE GROOVE (INCH POUNDS)

WHERE:

P = ALLOWABLE THRUST LOAD OF RINGS OR GROOVES (POUNDS)
T = RING THICKNESS (INCHES)
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)
d = GROOVE DEPTH (INCHES)
$_R$ = IMPACT CAPACITY OF RING (INCH POUNDS)

(g) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED =

WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON PAGES 10 AND 11 WILL APPLY. THE CORNER RADII AND CHAMFERS LISTED ON TABLE 1 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE VII. IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY, IN ACCORDANCE WITH THE FOLLOWING FORMULAS:

$P^1 = \frac{P\ CH}{CH^1}$

OR

$P^1 = \frac{PR}{R^1}$

WHERE:

P^1 = NEW ALLOWABLE THRUST LOAD
P = LISTED ALLOWABLE THRUST LOAD
CH^1 = NEW (SMALLER) CHAMFER
CH = LISTED CHAMFER
R^1 = NEW (SMALLER) CORNER
R = LISTED CORNER RADIUS

LIMIT LOADS LISTED IN TABLE VII ARE BASED ON RINGS OF BERYLLIUM COPPER (WORKING STRESS 180,000 PSI) FOR SIZES BELOW .250 INCH AND ON RINGS OF STEEL (WORKING STRESS 250,000 PSI) FOR SIZES .250 INCH AND OVER. IF THE ALLOWABLE GROOVE CAPACITY LOADS AS CALCULATED BY USING THE FORMULA ON PAGE 8 ARE LESS, THEN THEY SHOULD BE USED.

TABLE VII. Limit loads.

NOMINAL RING LOAD		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS	
FROM	TO	CARBON STEEL OR CRES	BERYLLIUM COPPER
.125	.156		45 LB
.188	.236		105 LB
.250	.469	470 LB	340 LB
.500	.672	910*LB	650 LB
.688	1.023	1340*LB	950 LB
1.062	1.500	1950 LB	1400 LB
1.562	2.000	3000 LB	2100 LB
2.062	2.688	5000 LB	
2.750	3.438	7350 LB	
3.500	5.000	10500 LB	
5.250	6.000	13500 LB	
6.250	7.000	21000 LB	
7.500	10.000	30000 LB	

4. MECHANICALLY DEPOSITED ZINC COATINGS (ASTM B695) FOR STEEL ARE PREFERRED. USING ELECTROPLATED ZINC (ASTM B633) HAS THE RISK OF HYDROGEN EMBRITTLEMENT OF HIGH STRENGTH STEEL, THEREFORE BAKING FOR HYDROGEN EMBRITTLEMENT PER ASTM B633 IS MANDATORY.

5. IN ACCORDANCE WITH A RECENTLY PUBLISHED A MEMORANDUM BY THE ASSISTANT SECRETARY OF THE ARMY FOR ACQUISITION EMPHASIZING THE REMOVAL OF HEXAVALENT CHROMIUM FROM ARMY EQUIPMENT, THE PURCHASE OR USE OF PARTS WITH CADMIUM COATING IS PROHIBITED WHEN ONE OF THE ALLOWED ZINC COATING WILL PROVIDE SUITABLE CORROSION PROTECTION.

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6. CHANGES FROM PREVIOUS ISSUE. THE MARGINS OF THIS SPECIFICATION ARE MARKED WITH VERTICAL LINES TO INDICATE WHERE CHANGES FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE PREVIOUS ISSUE

MILITARY INTEREST

Custodians:
Army – AR
Navy – OS
Air Force – 11
DLA – GS

Preparing activity:
DLA – GS6

Review activities:
Army – AT, AV, CR, GL, MI
Navy – AS, MC
Air Force – 11

(Project 5325-2022-018)

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